Figure 2.8 The red-green-refactor cycle of test-driven development

Furthermore, contract testing provides a fast feedback loop, enabling continuous refactoring and improvement of the codebase. With contract testing serving as a reliable and efficient validation mechanism, the team responsible for the data service can confidently refactor their codebase while ensuring compliance with the defined contracts. This fast feedback loop enhances code quality, promotes maintainability, and allows for effective evolution of the system.

2.3.3 Team benefits

Contract testing enforces various teams to communicate with each other, leading to improved project delivery and ensuring a shared understanding of requirements. By enforcing communication between consumer and provider teams, contract testing mitigates the risk of misinterpretations and misunderstandings that can arise during the development process. Contract testing enables teams to work more independently and autonomously.

One of the key aspects of contract testing is the creation of contracts in a clear and unambiguous manner. These contracts outline the expected interactions and outcomes between the consumer and provider services. They are typically written in plain English, using a standardized format such as the Given-When-Then (GWT) structure, known as the Gherkin syntax, which is popularized by the introduction of behavior-driven development (BDD). By having contracts written in a clear and accessible language, contract testing can serve as a source of truth and documentation for the entire team. The contract can also help with onboarding new team members, as well as ensuring continuity and consistency in development and testing practices.

2.4 Introducing contract testing to your teams

At this point, we hope you’re now convinced that contract testing is an important testing activity, especially when working with microservices architectures. Before introducing contract testing and a new tool to your team, there are a few things to consider because—let’s face it—this is an